1. a) $\left(600\,ns + \dfrac{x \times 10^9}{25 \times 1024} + 400ns\right) \times \dfrac{3072}{x}$

$= 120512 \times 10^3$ ns [$\because$ let, x bytes are transferred at a time]

$\Rightarrow 1000ns + 39062.5x$ ns

$= 39229.16667x$ ns

$\Rightarrow 1000ns = 166.6666667x$

$\therefore x = 6$

So, 6 bytes are transferred at a time in cycle stealing mode.

b) In burst mode:

$600ns + \dfrac{3072 \times 10^9}{25 \times 1024}$ ns + 400 ns

$= 120001000$ ns $= 120001$ μs

Here, 120001 μs < 120512 μs

So, burst mode is faster but it will create longest CPU wait period. That is why, I agree with Shafiq's statement.

2. a) I) ADD AX, BX → AX = AX + BX

AX :	6F24 H →	0110 (positive)	1111	0010	0100
(+) BX :	4213 H →	0100 (positive)	0010	0001	0011
	B137 H →	1011	0001	0011	0111

SF = 1 because MSB = 1

OF = 1 because positive + positive should be positive but the result is negative

b) SUB AX, BX → AX = AX − BX

AX :	B137 H →	1011 (negative)	0001	0011	0111
(−) BX :	4213 H →	0100 (positive)	0010	0001	0011
	6F24 H →	0110	1111	0010	0100

SF = 0 because MSB = 0

OF = 1 because negative - positive should be negative but the result is positive.

b) In memory mapped I/O memory is reduced by 50% because 1 bit of A.B. is used to determine the operation.

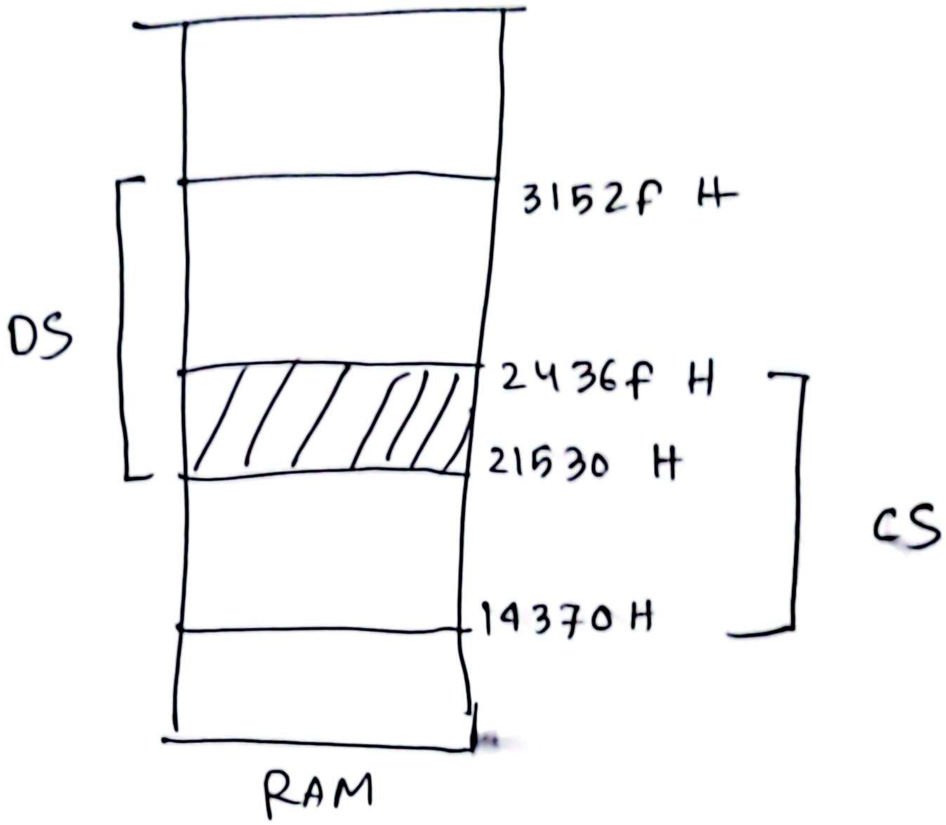
3. a) I) E.A. of CS = Segment * 10H + FFFF

= 1437 * 10H + FFFF

= 2436F H

II) E.A. of D.S. = 2153 * 10H + FFFF

= 3152F H



$\therefore$ Overlapping Segments = 2436F H

$- 21530 \text{ H} + 1$

$= 2E3F \text{ H} + 1$

$= \underline{2E40 \text{ H}}$

$\downarrow$

11840 slots

3b. $2^n * 24 = 2^{24} * 6$

$\Rightarrow 2^n = \frac{2^{24}}{4} = 2^{22}$

So, DRAM Addr. width = 22 bit

Memory Capacity

$= 2^{24} * 6$ bits

$= 2^{20} * 2^4 * 6$ bits

$= 96$ Mbits $= 12$ MB

5a) CS (LOW) stored at = 007EH

So, IP (LOW) is stored at = 007EH − 2H = 007CH

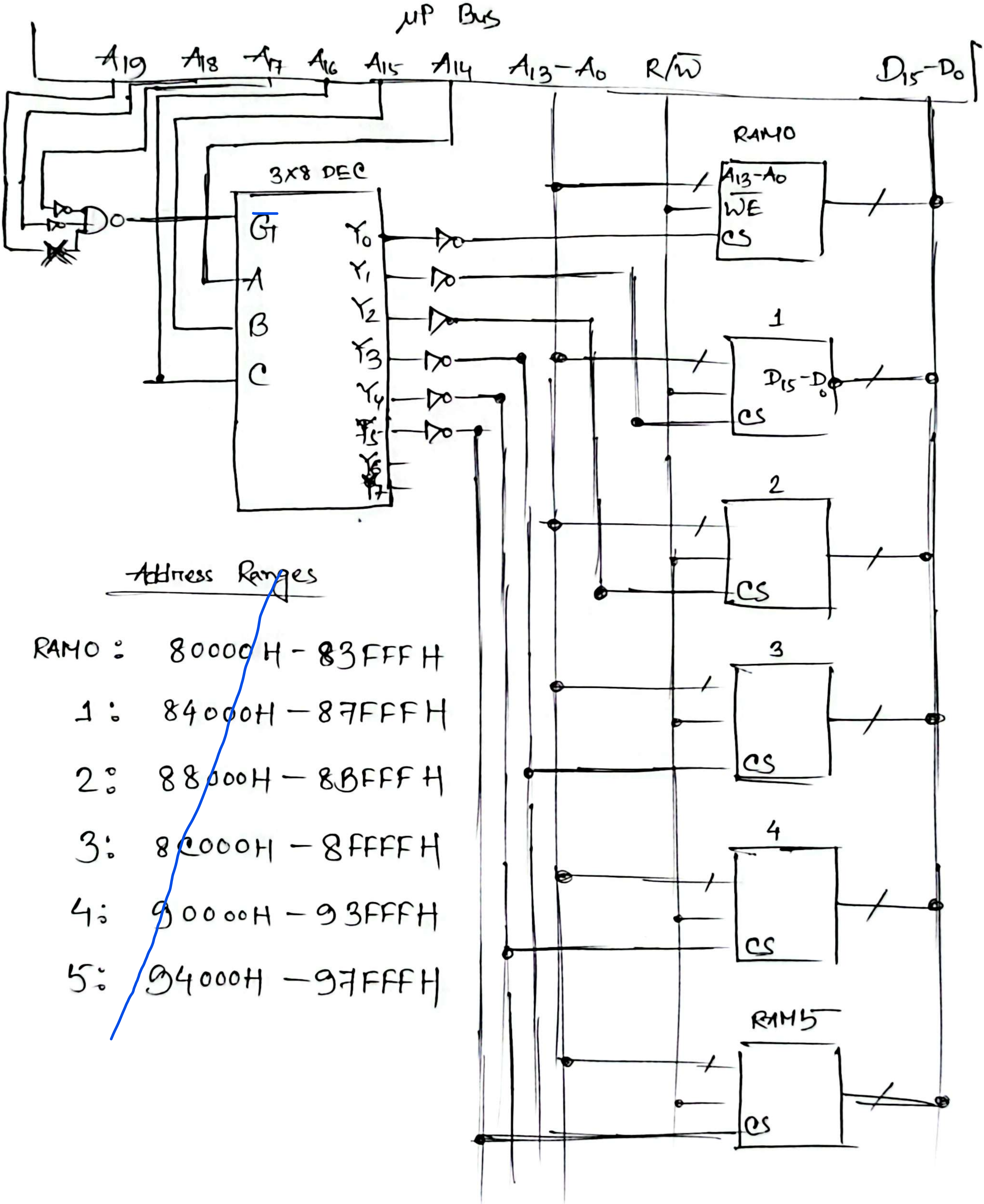
Hence the INT number = $\frac{007CH}{4H}$ = 1FH

5b)

1. Push FLAGS
2. Clear IF, TF
3. Push CS
4. Push IP
5. Load CS, IP
6. ISR
7. Pop IP
8. Pop CS
9. Pop FLAGS

(4a) RAM memory Capacity $= 2^{14} * 16$ bits $= 32KB$

\# RAMs required $= \frac{192}{32} = 6$



Address Ranges

RAM0 : 80000H - 83FFFH

1 : 84000H - 87FFFH

2 : 88000H - 8BFFFH

3 : 8C000H - 8FFFFH

4 : 90000H - 93FFFH

5 : 94000H - 97FFFH

4(a) ADDRESS MAPPINGS

RAM 0

Second :	80001 H
Sec-last:	83FFE H

RAM 1

Second :	84001 H
Sec-last:	87FFE H

RAM 2

Second :	88001 H
Sec-last:	8BFFE H